

1. Classical Cryptosystems

- **What are Classical Cryptosystems?**
 - Early methods of encryption used before modern computers.
 - Based on **substitution** (replacing characters) and **transposition** (rearranging characters).
 - Examples: Caesar Cipher, Vigenère Cipher, Playfair Cipher, etc.
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2. Substitution and Transposition Techniques

- **Substitution Techniques:**
 - Replace each character in the plaintext with another character.
 - Examples: Caesar Cipher, Monoalphabetic Cipher, Polyalphabetic Cipher.
 - **Transposition Techniques:**
 - Rearrange the characters in the plaintext to create ciphertext.
 - Examples: Rail Fence Cipher, Columnar Transposition.
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3. Caesar Cipher

- **What is the Caesar Cipher?**
 - A substitution cipher where each letter in the plaintext is shifted by a fixed number down the alphabet.
 - Example: With a shift of 3,
 - $A \rightarrow D, B \rightarrow E, C \rightarrow F, \dots, Z \rightarrow C.$
 - **Weakness:** Easy to break with brute force (only 25 possible shifts).

<https://www.youtube.com/watch?v=sMOZf4GN3oc>

4. Monoalphabetic Ciphers

- **What is a Monoalphabetic Cipher?**
 - A substitution cipher where each letter in the plaintext is mapped to a fixed different letter in the ciphertext.
 - Example:
 - $A \rightarrow Q, B \rightarrow W, C \rightarrow E, \dots, Z \rightarrow M.$

- **Weakness:** Vulnerable to frequency analysis (analyzing common letters like E, T, A).

<https://www.youtube.com/watch?v=1rJ4hHFdDyg&t=547s>

5. Polyalphabetic Ciphers

- **What is a Polyalphabetic Cipher?**
 - Uses multiple substitution alphabets to encrypt the plaintext.
 - Makes frequency analysis harder.
 - Example: **Vigenère Cipher**.
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6. Vigenère Cipher

- **What is the Vigenère Cipher?**
 - A polyalphabetic cipher that uses a keyword to shift letters in the plaintext.
 - Example:
 - Plaintext: HELLO
 - Keyword: KEY (K=10, E=4, Y=24)
 - Ciphertext: RIJVS
 - **Strength:** More secure than Caesar Cipher due to the use of a keyword.

Encryption

The plaintext(P) and key(K) are added modulo 26.

$$E_i = (P_i + K_i) \bmod 26$$

Decryption

$$D_i = (E_i - K_i) \bmod 26$$

<https://www.geeksforgeeks.org/vigenere-cipher/>
<https://www.youtube.com/watch?v=t58h3mPi6aM>

7. Playfair Cipher

- **What is the Playfair Cipher?**

- A digraph substitution cipher that encrypts pairs of letters (digraphs) instead of single letters.
- Uses a 5x5 grid filled with a keyword (excluding repeated letters and combining I/J).
- **Steps:**
 1. Break plaintext into digraphs (e.g., HE LL O).
 2. Apply rules based on the grid to encrypt each pair.
- **Strength:** More secure than simple substitution ciphers.

<https://www.geeksforgeeks.org/playfair-cipher-with-examples/>

<https://www.youtube.com/watch?v=zBDn5ATynqg>

8. Hill Cipher

- **What is the Hill Cipher?**

- A polygraphic substitution cipher that uses linear algebra (matrices) to encrypt blocks of letters.
- **Steps:**
 1. Convert letters to numbers (A=0, B=1, ..., Z=25).
 2. Use a key matrix to multiply and transform the plaintext into ciphertext.
- **Strength:** Resistant to frequency analysis.

<https://www.geeksforgeeks.org/hill-cipher/>

<https://www.youtube.com/watch?v=GY-psGvIKgA>

Finding Inverse: <https://byjus.com/maths/inverse-matrix/>

9. Vernam Cipher (One-Time Pad)

- **What is the Vernam Cipher?**

- A substitution cipher that uses a random key (as long as the plaintext) for encryption.
- Each key is used only once (One-Time Pad).
- **Steps:**
 1. Convert plaintext and key to binary.
 2. Perform XOR operation between plaintext and key to get ciphertext.
- **Strength:** Unbreakable if the key is truly random and used only once.

<https://www.geeksforgeeks.org/vernam-cipher-in-cryptography/>

<https://www.youtube.com/watch?v=UMPNIhSqJJY>

10. Rail Fence Cipher

- **What is the Rail Fence Cipher?**

- A transposition cipher that writes the plaintext in a zigzag pattern along "rails" and reads it row by row.
- Example:
 - Plaintext: HELLO
 - Rails: 2
 - Ciphertext: HLOEL
- **Weakness:** Easy to break with brute force.

<https://www.geeksforgeeks.org/rail-fence-cipher-encryption-decryption/>

<https://www.youtube.com/watch?v=MFwgaW>

https://www.tutorialspoint.com/cryptography/cryptography_rail_fence_cipher.htm

Summary Table of Classical Ciphers

Cipher	Type	Key Feature
Caesar Cipher	Substitution	Shifts letters by a fixed number.
Monoalphabetic Cipher	Substitution	Maps each letter to a fixed different letter.
Vigenère Cipher	Polyalphabetic	Uses a keyword to shift letters.
Playfair Cipher	Digraph Substitution	Encrypts pairs of letters using a 5x5 grid.
Hill Cipher	Polygraphic Substitution	Uses matrices to encrypt blocks of letters.
Vernam Cipher	Substitution	Uses a random key (One-Time Pad) for encryption.
Rail Fence Cipher	Transposition	Writes plaintext in a zigzag pattern and reads row by row.